Education and Income

Micro-Project 1

github.com/mascon0423/ANA500

Douglas Bobeck

September 20, 2026

Problem Statement

- This project looks at the relationship between education and income in the Adult dataset.
- Question: Is the share earning above $50,000 higher among adults with a bachelor's degree or higher than among adults with less education?
- Income is recorded in two groups: at or below $50,000 per year and above $50,000. The dataset does not give exact incomes.
- The work for this week is to load, explore, and clean the data so it can be used for a later comparison.

Hypothesis Formulation

- Hypothesis: Adults with at least a bachelor's degree are more likely to earn above $50,000 than adults with less education.
- Possible explanation: More education may help people qualify for jobs with higher pay. This explanation would need to be tested.
- The degree group includes Bachelors, Masters, Prof-school, and Doctorate. The comparison group includes all other education levels.
- Age, occupation, and hours worked may also affect income. This week's project prepares the data and leaves the hypothesis untested.

Acquire

- Source: UCI Adult dataset, created by Barry Becker and Ronny Kohavi from the 1994 Census database.
- Supplied file: adult income.csv. It contains 32,561 records and 15 columns: 14 attributes plus the income label.
- The supplied records match UCI's training split. The separate test split is not used in this micro-project.
- Tools used: Python in PyCharm, Pandas to load and organize the CSV as a DataFrame, and NumPy for numerical operations. The original CSV is kept unchanged.
- The main fields are education, education-num, and income. Age, occupation, and hours worked add context. fnlwgt is the final survey weight.
- Reference: Becker & Kohavi (1996), Adult. https://doi.org/10.24432/C5XW20

Prepare: Initial Exploration

- Income categories: 24,720 records at or below $50,000 and 7,841 above $50,000. These are sample counts.
- Missing workclass: 1,836. Missing occupation: 1,843. Missing native-country: 583.
- 2,399 records contain at least one missing value. The original marker is ?. Education and income have no missing values.
- 24 records repeat the same values as an earlier row. The file has no person ID, so these could represent different people.
- NumPy summaries: age ranges from 17 to 90, with median 37. Work hours range from 1 to 99 per week, with median 40.

Prepare: Cleaning Decisions

- Remove extra spaces and mark ? or blank cells as missing in Pandas. Leave unknown values missing instead of filling them with guesses.
- Change six numeric fields to integers and nine text fields to categorical data. Check that income labels are valid and education codes match their labels.
- Keep all records, including repeated rows. Record how many repeat so their effect can be checked in a later project.
- Keep valid zero capital gains and losses. Keep the observed age and work-hour values because they pass the range checks.
- Use NumPy to create education_group and income_gt_50k. The income flag is 1 for above $50,000 and 0 for at or below $50,000. Keep every original field.

Prepare: Final Dataset

- Prepared dataset: 32,561 records and 17 columns. No rows removed. Two derived fields added.
- Education groups: 8,067 bachelor's or higher, and 24,494 below bachelor's. Income rates by group have not been compared.
- Checks passed for column names, required values, numeric ranges, income labels, and matching education codes.
- Missing workclass, occupation, and native-country values remain recorded as missing. Education and income are complete for the planned comparison.
- These results describe the 1994 sample, with each row counted equally. They do not show that education causes higher income.
- This project ends at Step 2, Prepare. The hypothesis is still untested. Analyze, Report, and Act come later.